

NetSecure Project — Team Role Division Guide

This document defines the role division for a 3-person NetSecure cybersecurity project team. Each member has separate responsibilities while still understanding the overall workflow. The goal is to create a professional-level Network Vulnerability Scanner with a dashboard, risk detection system, and database integration.

Team Member	Primary Role	Main Focus
Person 1	Scanner Engineer + Project Lead	Networking, Port Scanning, Risk Logic
Person 2	Backend + Database Engineer	APIs, Flask, Database, Integration
Person 3	Dashboard + UI Engineer	Frontend, Charts, Visualization, UI

Person 1 — Scanner Engineer + Project Lead

Sections Covered:

- Section 1 — Networking + Port Scanner
- Section 2 — Service Detection
- Part of Section 3 — Risk Detection

Main Work:

- Build socket-based scanner
- Detect open ports
- Perform service detection
- Add threading for faster scans
- Create risk logic for insecure ports/services
- Manage GitHub integration and project coordination

Required Skills:

- Python basics
- Socket programming
- Networking fundamentals
- Git & GitHub basics

Main Aim:

Build the cybersecurity core of the project.

What They Achieve:

- Strong networking knowledge
- Cybersecurity fundamentals
- Real scanner-building experience
- Leadership and integration experience

Person 2 — Backend + Database Engineer

Sections Covered:

- Section 4 — Flask Backend
- Section 7 — Database Integration

Main Work:

- Build Flask APIs
- Connect scanner with frontend
- Store scan history in database
- Create backend routes
- Manage result processing
- Handle data flow between modules

Required Skills:

- Python basics
- Flask basics
- Database concepts
- API handling

Main Aim:

Create the system architecture and data management layer.

What They Achieve:

- Backend development skills
- API integration experience
- Database management knowledge
- Real software architecture understanding

Person 3 — Dashboard + UI Engineer

Sections Covered:

- Section 5 — Dashboard UI
- Section 8 — Charts & Visualization
- Section 9 — Project Polish

Main Work:

- Build dashboard interface
- Create scan forms
- Display scan results visually
- Add charts using Chart.js
- Improve UI/UX and responsiveness
- Create screenshots and README visuals

Required Skills:

- HTML/CSS basics
- JavaScript basics
- Frontend design principles
- Chart.js basics

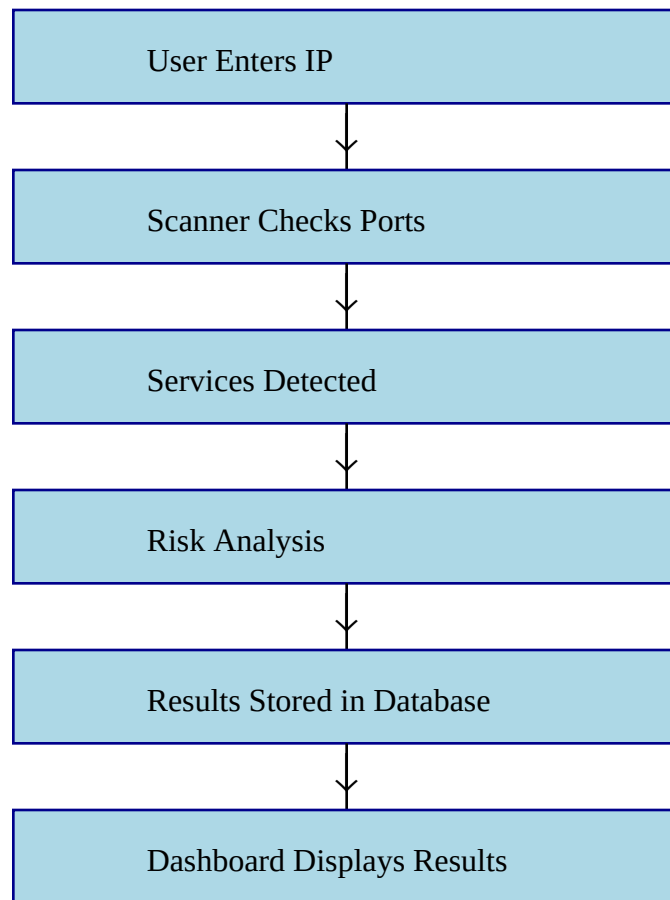
Main Aim:

Make the project visually professional and user-friendly.

What They Achieve:

- Frontend development experience
- Dashboard-building skills
- Data visualization knowledge
- UI/UX understanding

NetSecure Project Workflow



Final Team Outcome

- Fully functional Network Vulnerability Scanner
- Professional dashboard with charts and scan history
- Risk detection system for insecure services
- Real teamwork and GitHub collaboration experience
- Resume-worthy cybersecurity portfolio project
- Strong practical learning in networking, backend, frontend, and databases